

# Security in Cloud Computing

Module 38

Duration: 24:50

Video: <https://www.youtube.com/watch?v=5-6WRtoGJcQ>

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0:00 [Music]

0:20 Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture

0:28 we will study about Security in cloud computing first of all please see the contents which I shall

0:35 cover after covering brief introduction about Security in cloud computing I shall take up network security in cloud

0:44 and last I shall take up platform related security so now let us start

0:50 security is an important aspect to be considered in the cloud computing environment cloud computing security

0:59 refers to the C of procedures processes and standards designed to provide information security Assurance in an cloud computing

1:07 environment cloud computing security addresses both physical and logical security issues across all the different

1:15 service models of software platform and infrastructure it also addresses how these services are delivered public

1:24 private or hybrid delivery model Cloud security encompasses a broad broad range of security constraints from

1:31 an end user and Cloud provider's perspective where the end user will primarily will be concerned with the

1:39 provider security policy how and when their data is stored and who has access to that data subsequent sections talk

1:48 about data security virtualization security security issues in software as a service infrastructure as a service

1:58 and platform as a service model Etc security aspects cloud computing places business

2:06 data into the hands of an outside provider and makes Regulatory Compliance inherently riskier and more complex than

2:15 it is when systems are maintained in house loss of direct oversight means that the client company must verify that

2:24 the service provider is working to ensure that data security and integrity are IR cled the following are the

2:32 current security related research areas in cloud computing reliable distributed

2:39 applications based on the internet such as the e-commerce system rely heavily on the trust path among involved

2:47 parties the skyrocketing demand for a new generation of cloud-based consumer

2:54 and business applications is driving the need for the next generation of data centers that must be massively scalable

3:02 efficient agile reliable and secure in order to scale cloud services reliable to millions of service developers and

3:12 billions of end users the Next Generation cloud computing and data center infrastructure will have to

3:19 follow an evolution similar to the one that led to the creation of scalable telecommunication networks in the future

3:28 network-based cloud service providers will leverage virtualization Technologies to be able to allocate just

3:36 the right levels of virtualized compute Network and storage resources to individual applications based on

3:44 realtime business demand while also providing full service level Assurance of availability performance and security

3:52 at a reasonable cost data security due to huge infrastructure cost

4:00 organizations are slowly switching to Cloud technology data are stored in the cloud service provider infrastructure as

4:09 data do not reside in organization territory many complex challenges arise some of the complex data security

4:17 challenges in Cloud include the following cloud service models with multiple tenants sharing the same

4:26 infrastructure data mobility and legal issues relative to such government rules as European Union data privacy

4:36 directive lack of Standards about how cloud service providers securely recycle

4:42 disk space and erase existing data laws of visibility to key security and

4:49 operational intelligence that no longer is available to feed Enterprise it security intelligence and risk

4:58 management a new type of Insider who does not even work for your company but may have control and visibility into

5:06 your data data center security data are stored in outside territory of the user in a location

5:13 called as data center which is unknown to the user as the location of the data center is unknown to the user it becomes

5:21 a virtual data center the backbone of this virtual data center is virtual infrastructure or the virtual machine

5:30 however virtual platforms are dependent on many other often forgotten components of both the physical and virtual data

5:38 centers there are typically seven areas of concern that accompany any major virtual platform implementation or

5:46 migration often these issues are not seen during staging and testing and only appear when the virtual machines take on

5:55 the same amount of load as physical machines the critical points represent two Corner Stones of the data center

6:04 Network and storage lack of performance and availability virtualization moves many

6:12 input output tasks tuned for Hardware to software via the hypervisor the virtualization

6:19 translation layer is responsible for translating the optimized code for the software chip to the physical chip or

6:27 CPU running on the underlying Hardware lack of application awareness one of the limitation of

6:35 hypervisor and kernel based virtualization Solutions is that they only virtualize the operating system OS

6:44 virtualization does not virtualize nor is it even aware of applications that are running on the

6:51 OS even the same applications do not realize that they are using virtual Hardware on top of a

6:58 hypervisor overflowing storage Network although converting physical machines to Virtual machines is an asset

7:06 for building Dynamic data centers hard drives become extremely large flat file

7:13 virtual dis images consequently file storage becomes unmanageable congested

7:20 storage Network due to the portable nature of os virtualization there can be

7:26 Dramatical increase in data traversing the storage Network for the same reason that virtual machine disk files can

7:35 overrun physical storage once these images are made portable it becomes trivial to move these virtual machine

7:43 images across the network from one host to another or from one storage array to another network security cloud is based

7:53 on networking of many things together like the network of infrastructure while the network is the

7:59 backbone of the cloud many challenges are encountered in this network some of the challenges in the existing Cloud

8:07 networks are discussed in the following application performance Cloud tenant should be able to specify

8:15 bandwidth requirements for applications hosted in the cloud ensuring similar performance to the on premise

8:24 deployments many tiered applications requires some guaranteed bandwidth between server instances to satisfy user

8:32 transactions within an acceptable time frame and meet predefined service level

8:39 agreements insufficient bandwidth between these servers will impose significant latency on user

8:47 interactions flexible deployment of Appliance flexible deployment of

8:54 appliances Enterprises deploy a wide variety of security appliances in their data centers such as deep packet

9:04 inspection or intrusion detection systems and firewalls to protect their applications from attacks these are

9:13 often employed alongside other appliances that perform load balancing caching and application

9:20 acceleration when deployed in the cloud an Enterprise application should continue to be able to flexibly exploit

9:28 the functionality of these appliances policy enforcement complexities traffic isolation and

9:36 access control to end users are among the multiple forwarding policies that should be enforced these policies directly impact

9:46 the configuration of each router and switch changing requirements different

9:52 protocols open shortest path first link aggregation group Virtual Router

10:00 redundancy protocol vrrp and different flavors of L2 spanning tree protocols along with

10:10 vendor specific protocols make it extremely challenging to build operate and interconnect a cloud Network at

10:18 scale application rewriting applications should run out of the box as much as possible in

10:26 particular for IP addresses and Network dependent failover mechanisms applications may need to be

10:33 Rewritten or reconfigured before deployment in the cloud to address several Network related

10:41 limitations two key issues are first lack of an broadcast domain abstraction

10:47 in the cloud Network and second Cloud assigned IP addresses for virtual servers location

10:56 dependency Network appliances and servers are typically tied to a statically

configured physical Network

- 11:04 which implicitly creates a location dependent constraint for instance the IP address of a server is typically
- 11:12 determined based on the vcn or the subnet to which it belongs V lens and subnets are based on the physical switch
- 11:20 Port configuration therefore a virtual machine cannot be easily and smoothly migrated across the network constr
- 11:29 virtual machine migration decreases the level of resource utilization and flexibility besides physical mapping of
- 11:38 Wheeland or subnet space to the physical ports of a switch often leads to a fragmented IP address
- 11:45 pool platform related security cloud service providers offer services in various service models like
- 11:54 s AAS Bas and I a as where the user is offered
- 12:02 varied Services based on his or her requirements every service model offer brings with it many security related
- 12:10 challenges like secured Network locality of resources accessing secure data data
- 12:17 privacy and backup policy security issues in cloud service models cloud computing uses Three Delivery models
- 12:25 such as s a p AA S and L AAS through which different types of Computing
- 12:34 Services are provided to the end user these three delivery models provide infrastructure resources application
- 12:41 platform and software as services to the cloud customer these service models plays a different level of security
- 12:49 requirements in the cloud environment L is the basis of all cloud services
- 12:57 with paa s built upon it and s in turn
- 13:03 built upon it just as capabilities are inherited so are the information security issues and risks there are
- 13:13 significant trade-offs to each model in terms of integrated features complexity versus extensibility and
- 13:21 security if the cloud service provider takes care of only the security at the lower part of the security architecture
- 13:29 the consumers become more responsible for implementing and managing their security capabilities
- 13:37 saas is a softwar deployed model in which applications are remotely hosted
- 13:44 by the application or service provider and made available to customers on demand over the Internet the saas model
- 13:54 offers the customers with significant benefits such as improved op operational efficiency and reduced costs

- 14:03 saas is rapidly emerging as the dominant delivery model for meeting the needs of Enterprise IT
- 14:11 services saas is rapidly emerging as the dominant delivery model for meeting the needs of
- 14:20 Enterprise IT services p is one layer above
- 14:26 L on the stack and racts a everything up to OS middleware Etc it offers
- 14:34 developers a service that provides a complete software development life cycle management from planning to design to
- 14:42 building applications to deployment to testing to maintenance everything else is abstract
- 14:49 away from the view of the developers software as a service security issues in a traditional on
- 14:58 premise application deployment model the sensitive data of each Enterprise continue to reside within the Enterprise
- 15:06 boundary and are subject to its physical logical and personal security and exess control policies
- 15:14 SAA applications are accessed through the web and so web browser security is very much
- 15:22 important information security officers will need to consider various methods of securing SA a
- 15:30 applications web services security extensible markup language encryption
- 15:36 SSL and available options used in enforcing data protection transmitted over the Internet this involves the use
- 15:45 of strong encryption techniques for data security and F grained authorization to
- 15:51 control access to data the pain points of concern in saas are as follows network security in
- 16:01 16 minutes, 1 second an saas deployment model sensitive data flow over the network needs to be
- 16:08 secured in order to prevent leakage of sensitive information this involves the use of strong Network traffic encryption
- 16:16 techniques such as the SSL and TLS for security resource
- 16:23 locality in an saas model of a cloud environment the end users use the
- 16:30 services provided by the cloud providers without knowing exact where the resources for such services are located
- 16:38 due to the compliance and data privacy rules in various countries locality of data is of utmost importance in much
- 16:47 Enterprise architecture Cloud standards to achieve interoperability among clouds and to
- 16:55 increase their stability and security Cloud standard s are needed across

organizations for example the current

17:03 storage Services by a cloud provider may be incompatible with those for other providers in order to keep their

17:11 customers Cloud providers May introduce socalled sticky services that create difficulty for the users if they want to

17:19 migrate from one provider to the other data Access Data access issue is mainly

17:26 related to security policies provide to the users while accessing the data the organizations will have their

17:34 own security policies based on which each employee can have access to a particular set of data the security

17:41 policies May entitle some considerations wherein some of the employees are not given access to a certain amount of data

17:50 these security policies must be ader by the cloud to avoid intrusion of data by unauthorized users

17:59 the saas model must be flexible enough to incorporate these specific policies put

18:06 forward by the organization the model must also be able to provide organizational boundary within the cloud because multiple

18:15 organizations will be deploying their business processes within a single Cloud environment data

18:22 breaches since data from various users and business organizations lie together in a cloud environment breaching into

18:31 the cloud environment will potentially attack the data of all the users thus the cloud becomes a high value Target

18:41 backup the saas vendor needs to ensure that all sensitive Enterprise data are

18:48 regularly backed up to facilitate quick recovery in case of disasters also the use of strong encryption schemes to

18:56 protect the backup data is recommended to prevent accidental leakage of sensitive information in the case of

19:03 cloud vendors such as Amazon the data at RIS in est3 are not encrypted by default

19:11 the users need to separately encrypt their data and backups so that it cannot be accessed or tampered with by

19:19 unauthorized parties platform as a service security issues

19:27 paas Pro provides a ready to use platform including OS that runs on vendor provided

19:34 infrastructure as the infrastructure is of the cloud service provider various security challenges of the focused

19:41 architecture are caused mainly by the spread of the user objects over the hosts of the cloud stringently allowing

19:50 excess of objects to the resources and defending the objects against malicious or

corrupt providers reasonably reduce

19:58 possible risks network access and service measurement bring together concerns about secure Communications and excess

20:08 control well-known practices object scale enforcement of authorization and

20:14 undeniable traceability methods May alleviate the concerns apart from the aform problems

20:22 user privacy must be protected in a public share Cloud therefore prop most solutions must be privacy aware service

20:32 continuity is another concern for many Enterprises that consider Cloud adoption accordingly fall tolerant reliable

20:41 systems are required infrastructure as a service security issues cloud computing

20:48 makes a lot of promises in the areas of increased flexibility and Agility potential cost savings and

20:56 competition advantages for for developers so that they can stand up and infrastructure quickly and efficiently to enable them to develop the software

21:05 to drive business success there are a lot of problems that cloud especially private Cloud solves but it is not that

21:14 much good is solving problems related to security however in a private Cloud environment some of the traditional

21:23 problems faced are as follows hypervisor security in private Cloud most or all of the services will

21:32 run in a virtualized environment and the security model used by the hypervisor cannot be taken for granted a need to

21:41 evaluate the security models and the development of hypervisors becomes

21:48 necessary multi- tency although all the tenants in the multi- tency environment will be from

21:56 the same company not all rents may be comfortable sharing infrastructure with other users within the same company

22:04 identity management and access control in our traditional data center we were comfortable with the small handful of

22:12 authentication repositories we had to work with active directory being one of the most popular but with private Cloud

22:21 handling authentication and authorization for the cloud infrastructure handling denance and handling delegation of administration of

22:30 various aspects of the cloud fabric are the major task to be addressed network security in private Cloud we are likely

22:39 to have many components of a service communicate with each other over virtual network channels only accessing the traffic employing

22:49 some powerful access controls for physical networks and control quality of service which is a key issue in the

- 22:56 availability aspect of the confidentiality integrity and availability security model are major
- 23:05 concerns combining the three types of clouds public Cloud private cloud and hybrid Cloud together with the three
- 23:14 Service delivery models we get a complete picture of a cloud computing environment interl by connectivity
- 23:22 devices coupled with information security components virtualized physical sources
- 23:29 virtualized infrastructure virtualized middleware platforms and business related applications are being provided
- 23:37 as Computing Services in the cloud cloud providers and Cloud consumers must be able to maintain and
- 23:46 establish Computing security at all levels of interfaces in the cloud computing architecture so friends this was all
- 23:55 about Security in cloud computing hope the concepts explained in this lecture were understandable and helpful hope to
- 24:03 see you in next lecture till then goodbye thank you
- 24:08 [Music]
- 24:46 [Music]